Cured and broiled meat consumption in relation to childhood cancer: Denver, Colorado (United States)

The association between cured and broiled meat consumption by the mother during pregnancy and by the child was examined in relation to childhood cancer. Five meat groups (ham, bacon, or sausage; hot dogs; hamburgers; bologna, pastrami, corned beef, salami, or lunch meat; charcoal broiled foods) were assessed. Exposures among 234 cancer cases (including 56 acute lymphocytic leukemia [ALL], 45 brain tumor) and 206 controls selected by random-digit dialing in the Denver, Colorado (United States) standard metropolitan statistical area were compared, with adjustment for confounders. Maternal hot-dog consumption of one or more times per week was associated with childhood brain tumors (odds ratio [OR] = 2.3, 95 percent confidence interval [CI] = 1.0-5.4). Among children, eating hamburgers one or more times per week was associated with risk of ALL (OR = 2.0, CI = 0.9-4.6) and eating hot dogs one or more times per week was associated with brain tumors (OR = 2.1, CI = 0.7-6.1). Among children, the combination of no vitamins and eating meats was associated more strongly with both ALL and brain cancer than either no vitamins or meat consumption alone, producing ORs of two to seven. The results linking hot dogs and brain tumors (replicating an earlier study) and the apparent synergism between no vitamins and meat consumption suggest a possible adverse effect of dietary nitrites and nitrosamines.